

A DOUBLE-SUSPENSION COUNTEREXAMPLE TO THE COMPOSITION CONVERSE FOR STRONG CW-REGULAR SUBDIVISIONS

ABSTRACT. Question 8.6 of Stapledon’s *Subdivisions of lower Eulerian posets* (arXiv:2511.16608v1) asks whether the converse of his Lemma 7.15 survives the removal of the hypothesis $\text{rank}(\sigma) = 0$: if $\sigma: X \rightarrow Y$ and $\tau: Y \rightarrow Z$ are strong formal subdivisions of lower Eulerian posets and both σ and $\tau \circ \sigma$ are strong CW-regular subdivisions, must τ be one? We answer **no** with one finite triple, of ranks $\text{rank}(\sigma) = 2$ and $\text{rank}(\tau) = 4$. Let P be the 16-vertex triangulation of the Poincaré homology 3-sphere of Björner and Lutz, $Y = \text{face}(P)$ with the natural rank shifted by 2, $Z = B_0$ with $\rho_Z(\hat{0}_Z) = 6$, $X = \text{face}(P * \partial\Delta^2)$ with its natural rank, σ the join projection and τ the unique map. Then $|K_X|$ is the double suspension of $|P|$, homeomorphic to S^5 by Cannon’s theorem, so $\tau \circ \sigma$ is strong CW-regular; σ is strong CW-regular by the join-of-balls example of the same paper; and τ is not, because $|P|$ is not a sphere. Nothing beyond this single triple is claimed.

1. THE QUESTION

The last of the five questions of Section 8 of Stapledon [4], *Generalizations and further questions*, reads as follows.

Question 8.6 of [4]. Does the converse statement in Lemma 7.15 hold without the assumption that σ has rank 0 (c.f. Lemma 3.14)? That is, let $\sigma: X \rightarrow Y$ and $\tau: Y \rightarrow Z$ be strong formal subdivisions between lower Eulerian posets X, Y, Z with rank functions ρ_X, ρ_Y, ρ_Z respectively. Assume that σ and $\tau \circ \sigma$ are strong CW-regular subdivisions. Is τ necessarily a strong CW-regular subdivision?

The question is universally quantified over triples, so a single triple decides it. Four conventions of [4] are load-bearing.

- (D1) Definition 7.2: “We say that K is a regular CW-sphere or regular CW-ball if $|K|$ is *homeomorphic* to $S^{\dim K}$ or $D^{\dim K}$ respectively.” Homeomorphism only — not piecewise linear, not shellable.
- (D2) Definition 7.11: an order-preserving, rank-increasing, strongly surjective $\sigma: X \rightarrow Y$ of CW-posets is a *strong CW-regular subdivision* if for every $y \in Y$,

$$\dim K_{X, \leq y} = \rho_Y(y) - \rho_X(\hat{0}_X) - 1,$$

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- and $K_{X, \leq y}$ is a regular CW-sphere when $y = \hat{0}_Y$, and a regular CW-ball with boundary $K_{X, < y}$ otherwise; here $X_{\leq y} = \sigma^{-1}([\hat{0}_Y, y])$.
- (D3) With the conventions of [4, §7.1], B_0 is a CW-poset, equal to the face poset of the empty complex, and $\text{face}(K)$ carries the natural rank $\rho(e) = \dim e + 1$; the rank of a strong CW-regular subdivision is $\text{rank}(\sigma) = \rho_Y(\hat{0}_Y) - \rho_X(\hat{0}_X)$ ([4, §7.2]).
- (D4) In Lemma 7.15 the forward implication holds unconditionally, and the converse is proved *only* under the extra hypothesis $\text{rank}(\sigma) = 0$. It is stated for CW-posets, which is stronger than the “lower Eulerian” of Question 8.6, so a CW-poset witness refutes both readings.

2. THE WITNESS

Let P be the 16-vertex triangulation of the Poincaré homology 3-sphere $\Sigma(2, 3, 5)$ of Björner and Lutz [1], listed by its 90 tetrahedra in Table 1. It has $f(P) = (16, 106, 180, 90)$ and $\dim P = 3$.

1 2 4 9	1 2 4 15	1 2 6 14	1 2 6 15	1 2 9 14
1 3 4 12	1 3 4 15	1 3 7 10	1 3 7 12	1 3 10 15
1 4 9 12	1 5 6 13	1 5 6 14	1 5 8 11	1 5 8 13
1 5 11 14	1 6 13 15	1 7 8 10	1 7 8 11	1 7 11 12
1 8 10 13	1 9 11 12	1 9 11 14	1 10 13 15	2 3 5 10
2 3 5 11	2 3 7 10	2 3 7 13	2 3 11 13	2 4 9 13
2 4 11 13	2 4 11 15	2 5 8 11	2 5 8 12	2 5 10 12
2 6 10 12	2 6 10 14	2 6 12 15	2 7 9 13	2 7 9 14
2 7 10 14	2 8 11 15	2 8 12 15	3 4 5 14	3 4 5 15
3 4 12 14	3 5 10 15	3 5 11 14	3 7 12 13	3 11 13 14
3 12 13 14	4 5 6 7	4 5 6 14	4 5 7 15	4 6 7 11
4 6 10 11	4 6 10 14	4 7 11 15	4 8 9 12	4 8 9 13
4 8 10 13	4 8 10 14	4 8 12 14	4 10 11 13	5 6 7 13
5 7 9 13	5 7 9 15	5 8 9 12	5 8 9 13	5 9 10 12
5 9 10 15	6 7 11 12	6 7 12 13	6 10 11 12	6 12 13 15
7 8 10 14	7 8 11 15	7 8 14 15	7 9 14 15	8 12 14 15
9 10 11 12	9 10 11 16	9 10 15 16	9 11 14 16	9 14 15 16
10 11 13 16	10 13 15 16	11 13 14 16	12 13 14 15	13 14 15 16

TABLE 1. The 90 tetrahedra of P , the 16-vertex triangulation of the Poincaré homology 3-sphere [1]. The properties of P that the proof below consumes — its f -vector, that it is a pure closed 3-manifold, that $H_*(P) \cong H_*(S^3)$ over $\mathbb{Z}$, and that $|P|$ is not simply connected — are checked as described in §4.

Put

$$\begin{aligned} Y &= \text{face}(P), & \rho_Y &= (\dim + 1) + 2, \text{ so } \rho_Y(\widehat{0}_Y) = 2, \\ Z &= B_0 = \text{face}(\emptyset), & \rho_Z(\widehat{0}_Z) &= 6, \\ K_X &= P * \partial\Delta^2, & \dim K_X &= 3 + 1 + 1 = 5, \\ X &= \text{face}(K_X) = Y \times \partial B_3, & \rho_X &= \dim + 1, \text{ so } \rho_X(\widehat{0}_X) = 0, \end{aligned}$$

with cardinalities $|Y| = 393$, $|Z| = 1$ and $|X| = 7 \cdot 393 = 2751$, with $\sigma: X \rightarrow Y$ the join projection $(F, G) \mapsto F$ and $\tau: Y \rightarrow Z$ the unique map. The shift of ρ_Y by $+2$ is licensed by [4, Example 3.4]. Then

$$\text{rank}(\sigma) = 2 - 0 = 2, \quad \text{rank}(\tau) = 6 - 2 = 4, \quad \text{rank}(\tau \circ \sigma) = 6 - 0 = 6 = 2 + 4,$$

and the two routes to $\rho_Z(\widehat{0}_Z)$ agree: $0 + 6 = 6$ and $2 + 4 = 6$. Since $\text{rank}(\sigma) = 2 > 0$ the triple lies outside the half of Lemma 7.15 that is proved, so it answers Question 8.6 and contradicts nothing published.

Theorem 1. *For the triple above, σ and τ are strong formal subdivisions, σ and $\tau \circ \sigma$ are strong CW-regular subdivisions, and τ is not. Hence the answer to Question 8.6 is **no**.*

Proof. σ is order-preserving, rank-increasing and strongly surjective. It is a projection of a product order, hence order-preserving; $\rho_X((y, z)) = (\dim e_y + 1) + (\dim e_z + 1) \leq \rho_Y^{\text{nat}}(y) + 2 = \rho_Y(\sigma(y, z))$ because $\rho_{\partial B_3}(z) \in \{0, 1, 2\}$; and given (F, G) and $y \geq F$ one takes (y, G') with G' an edge of $\partial\Delta^2$ containing G , whose rank is $|y| + 2 = \rho_Y(y)$, which is strong surjectivity ([4, Definition 3.1]). For τ , strong surjectivity says every face of P lies in a tetrahedron, and P is pure, being a triangulated closed 3-manifold.

(i) *Both maps are strong formal subdivisions*, verified directly from [4, Definition 3.5] rather than deduced from CW-regularity, which would be circular because (D2) presupposes it. For σ , the summand $(-1)^{\rho_Y(y) - \rho_X(x')}$ over $x' = (y, G') \geq (F, G)$ with $\sigma(x') = y$ reduces to $(-1)^{|G'|}$, and the sum over $G' \supseteq G$ inside $\text{face}(\partial\Delta^2)$ is $1 - 3 + 3 = 1$ for $G = \emptyset$, $-1 + 2 = 1$ for G a vertex and $+1 = 1$ for G an edge. For τ the summand reduces to $(-1)^{|F'|}$, giving $1 - 16 + 106 - 180 + 90 = 1$ for $F = \emptyset$ and, for $|F| = k \geq 1$ with $\text{lk}(F) \cong S^{3-k}$, $(-1)^k(-1)^{4-k} = +1$. All five sums equal 1. The τ half is exactly [4, Example 7.17] at $r = 4$ with $K = P$.

(ii) σ is a strong CW-regular subdivision, the only input being the ball structure of [4, Example 7.10], which we quote. Since $X_{\leq y} = [\widehat{0}_Y, y] \times \partial B_3$, we have $K_{X, \leq y} = \overline{e}_y * \partial\Delta^2$, of dimension $\dim e_y + 2$, and the requirement of (D2) is $\rho_Y(y) - \rho_X(\widehat{0}_X) - 1 = (\dim e_y + 3) - 0 - 1 = \dim e_y + 2$: equal for every y , including $y = \widehat{0}_Y$, where $-1 + 2 = 1 = \dim \partial\Delta^2$. At $y = \widehat{0}_Y$ one has $K_{X, \leq \widehat{0}_Y} = \partial\Delta^2$, a triangle boundary, whose space is S^1 : a regular CW-sphere. For $y \neq \widehat{0}_Y$, $\overline{e}_y$ is a simplex Δ^d and [4, Example 7.10] makes $\Delta^d * \partial\Delta^2$ a ball of dimension $d + 2$ whose boundary is $\partial\Delta^d * \partial\Delta^2$, which is precisely $K_{X, < y}$ because $X_{< y} = [\widehat{0}_Y, y) \times \partial B_3$.

(iii) $\tau \circ \sigma$ is a strong CW-regular subdivision, of rank 6. The only element of Z is $\widehat{0}_Z$, so $K_{X, \leq \widehat{0}_Z} = K_X$, and the dimension requirement is $6 - 0 - 1 = 5 = \dim K_X$. For the sphere clause, the join of the underlying spaces gives

$$|K_X| = |P| * |\partial \Delta^2| \cong |P| * S^1 \cong |P| * S^0 * S^0 \cong \Sigma^2 |P|,$$

using the standard identifications $|\partial \Delta^2| \cong S^1 \cong S^0 * S^0$, the associativity of the join of compact spaces, and $\Sigma W \cong S^0 * W$; so $|K_X|$ is the double suspension of $|P|$. Now P is a closed 3-manifold with $H_*(P) \cong H_*(S^3)$ over $\mathbb{Z}$ — both checked as described in §4 — which is the homology-3-sphere hypothesis under which we invoke Cannon’s double suspension theorem [2] (see also [3]); it gives $\Sigma^2 |P| \cong S^5$. By (D1) only a homeomorphism is asked for, so K_X is a regular CW-sphere.

(iv) τ is not a strong CW-regular subdivision. Again the only element of Z is $\widehat{0}_Z$ and $K_{Y, \leq \widehat{0}_Z} = P$. The dimension clause *passes*: $\rho_Z(\widehat{0}_Z) - \rho_Y(\widehat{0}_Y) - 1 = 6 - 2 - 1 = 3 = \dim P$, so the failure is not a bookkeeping artefact. The sphere clause fails, because $\pi_1(|P|) \neq 1$ by Proposition 2 below and a space homeomorphic to S^3 is simply connected. Equivalently, by [4, Example 7.16] a strong CW-regular subdivision onto B_0 forces K_Y to be a regular CW-sphere.

All hypotheses of Question 8.6 hold and its conclusion fails. $\square$

3. THAT $|P|$ IS NOT A SPHERE, EXHIBITED

Clause (iv) is the only place where the failure lives, and it rests on a single negative fact about P , which is exhibitable on the page.

Let A_5 be the alternating group on $\{1, \dots, 5\}$, written multiplicatively with left-to-right composition, so that $(gh)(i) = h(g(i))$ and the product of the labels along an edge path is read left to right. Table 2 assigns to each ordered edge uv of P with $u < v$ an element $g_{uv} \in A_5$, written as the one-line word $g(1)g(2)g(3)g(4)g(5)$; every edge of P absent from the table carries the identity, and $g_{vu} := g_{uv}^{-1}$.

Proposition 2. *The labelling of Table 2 satisfies $g_{uv}g_{vw} = g_{uw}$ at every one of the 180 triangles $\{u < v < w\}$ of P . Hence holonomy is a well-defined homomorphism $h: \pi_1(|P|) \rightarrow A_5$; its image is all of A_5 , and the closed edge path $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ has holonomy $g_{12}g_{23}g_{13}^{-1} = (1\,2\,4\,5\,3) \neq 1$. In particular $\pi_1(|P|) \neq 1$ and $|P|$ is not homeomorphic to S^3 , even though $H_*(P) \cong H_*(S^3)$ with no torsion.*

Proof. π_1 of a simplicial complex is the edge-path group of its 2-skeleton, generated by the edges outside a spanning tree and presented by one relation $g_{uv}g_{vw} = g_{uw}$ per triangle. A labelling satisfying every triangle relation is therefore exactly a homomorphism to A_5 once the tree labels are the identity, which they are; the holonomy of the loop obtained by closing an edge uv through the tree is g_{uv} , so the image of h is the subgroup generated by the labels. The three edges 12, 23, 13 all lie in P while $\{1, 2, 3\}$ is *not* a

23: 24153	25: 35214	27: 24153	28: 35214
210: 24153	211: 35214	212: 43521	213: 41532
35: 23451	311: 23451	313: 54213	314: 23451
45: 23451	46: 23451	47: 35214	48: 41532
410: 41532	411: 35214	413: 41532	414: 23451
57: 43521	59: 25413	510: 51234	512: 25413
515: 51234	67: 43521	610: 24153	611: 43521
612: 43521	79: 31524	713: 54213	714: 31524
715: 43152	89: 25413	812: 25413	814: 31524
815: 43152	910: 35421	913: 41532	915: 35421
1011: 54132	1012: 54132	1014: 31524	1016: 54132
1113: 35421	1115: 43152	1213: 54213	1214: 23451
1215: 54213	1314: 54132	1316: 54132	1415: 35421
1516: 54132			

TABLE 2. An A_5 -valued flat connection on the 2-skeleton of P : the 53 edges of P whose label is not the identity, each label a 5-cycle written as a one-line word. The remaining 53 edges carry the identity, and they contain a spanning tree of the 1-skeleton: for instance the 14 edges of the star of the vertex 1 together with 1416, which is why the image of the holonomy homomorphism is generated by the labels themselves.

face of P — which is why its holonomy is not forced to be trivial — and $g_{12} = g_{13} = 1$ while $g_{23} = (1\ 2\ 4\ 5\ 3)$. The 180 relations, the generated image and the homology of P are checked as described in §4; every one of them is a finite computation in exact arithmetic on Table 1 and Table 2. $\square$

4. VERIFICATION

The two objects the argument consumes are printed above in full: the 90 tetrahedra of Table 1 and the 53 labels of Table 2. The accompanying program `verify.py` (Python 3.9+, standard library only, exact integer arithmetic, no external data) transcribes exactly those two tables and re-derives the finite quantities claimed here: $f(P) = (16, 106, 180, 90)$; that every triangle of P lies in exactly two tetrahedra, every vertex link is a combinatorial 2-sphere and every edge link a single cycle, so P is a closed 3-manifold; that $H_*(P) = (\mathbb{Z}, 0, 0, \mathbb{Z})$ with no torsion, by Smith normal form over $\mathbb{Z}$; the 180 triangle relations of Proposition 2, that the labels generate a group of order 60, and the nontrivial holonomy of $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$; $f(K_X) = (19, 157, 546, 948, 810, 270)$ and $|X| = 2751 = 393 \cdot 7$; the rank arithmetic $2 + 4 = 6$ from both sides; that σ is order-preserving, rank-increasing and strongly surjective; all 23,359 alternating sums of [4, Definition 3.5] for σ and all 393 for τ , each equal to 1; and the dimension clause of (D2) at every one of the 393 elements of Y together with the boundary

bookkeeping at the 392 elements $y \neq \widehat{0}_Y$. The program reports **VERDICT: ALL 52 CHECKS PASS**.

Three inputs to the proof are quoted rather than reproved, and the program does not touch them: Cannon's theorem, and with it every homeomorphism type, none of which is computed anywhere; the ball structure of [4, Example 7.10], of which only the combinatorial consequences are checked; and the classical identification of $\pi_1(|P|)$, of which only the surjection onto A_5 is proved and only that is needed. This paper does not address the case $\text{rank}(\sigma) = 1$ of Question 8.6; it claims no minimality, and it makes no claim about other complexes in place of P , about other ranks, or about the literature, which the program does not search.

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